Feature Importance and Model Performance Analysis

Introduction

In this analysis, I evaluated the importance of various features and compared the performance of different machine learning models for predicting across a variety of features and time windows. To assess feature importance, I employed a method where each feature was removed one at a time, and the model's performance was measured using mean absolute error (MAE).

Experiment Configurations

During my experimentation, I tested Linear, Dense, CNN, RNN, LSTM, Bi-LSTM, Autoregressive, and ARIMA models on the following configurations:

Feature: SWC_5 (Soil Water Content at 5 cm depth)

- Input steps: 24, Output steps: 1
- Input steps: 24, Output steps: 6
- Input steps: 48, Output steps: 12
- Input steps: 168 (7 days), Output steps: 24
- Input steps: 168 (7 days), Output steps: 48

Feature: SWC_10 (Soil Water Content at 10 cm depth)

- Input steps: 24, Output steps: 1
- Input steps: 24, Output steps: 6
- Input steps: 48, Output steps: 12
- Input steps: 168 (7 days), Output steps: 24
- Input steps: 168 (7 days), Output steps: 48

Feature: SWC_20 (Soil Water Content at 20 cm depth)

- Input steps: 24, Output steps: 1
- Input steps: 24, Output steps: 6
- Input steps: 48, Output steps: 12

- Input steps: 168 (7 days), Output steps: 24
- Input steps: 168 (7 days), Output steps: 48

Feature Importance Results

Initial feature importance experimentation was conducted on a CNN model handling an input window of 1 week of hourly data to predict the next 48 hours of *SWC_5*. The feature importance results indicated that the feature *T_50* was the most crucial, as removing it resulted in the highest MAE of 0.2715. This suggests that soil temperature at 50 cm depth plays a significant role in the model's predictions. In contrast, the feature *Wx* had the lowest impact, with an MAE of 0.2155 when removed, implying that it is less critical to the model's accuracy.

More feature importance experimentation is being done to determine how features affect the different models previously analyzed and when utilizing different model configurations. The configurations being analyzed are as follows:

- **Feature: SWC_5** *using 48 hours of input to predict 12 hours*
- **Feature: SWC_10** *using 48 hours of input to predict 12 hours*
- **Feature: SWC_20** *using 48 hours of input to predict 12 hours*

Model Performance Findings

When evaluating model performance, the **multi-step linear model** consistently outperformed the other machine learning models across most configurations, achieving the lowest MAE in nearly every scenario. In configurations with shorter input and output sequences, such as the **SWC_5** and **SWC_10** setups, the linear model consistently delivered the lowest MAE, with values as low as **0.0084** in the **SWC_20 - 24 input / 1 output** configuration. Conversely, **baseline models** and

autoregressive models exhibited significantly higher error rates, particularly in longer input/output configurations, with MAE values reaching as high as **0.4215**.

Models like **CNN**, **RNN**, and **LSTM** performed well in several cases, particularly with larger input sequences, but none consistently outperformed the multi-step linear model. Notably, the **RNN** and **Bi-LSTM** models provided competitive results but showed less consistency across configurations. The **baseline model** showed the highest MAE in many configurations, confirming that more complex deep learning models like RNN and CNN offer better suitability for this soil moisture content forecasting task. Overall, the results confirm that multi-step linear and deep learning models are more effective, with **linear models** excelling in most scenarios.

Full Results

Model Performance Summary Across All Configurations:

Configuration: SWC_5 - 24 input / 1 output
Model: Baseline - Loss: 0.0046, MAE: 0.0149
Model: Multi-step Linear - Loss: 0.0646, MAE: 0.2153
Model: Multi-step Dense - Loss: 0.0060, MAE: 0.0507
Model: CNN - Loss: 0.0079, MAE: 0.0584
Model: RNN - Loss: 0.0034, MAE: 0.0253
Model: LSTM - Loss: 0.0110, MAE: 0.0788
Model: Autoregressive - Loss: 0.0370, MAE: 0.1031
Model: Bi-LSTM - Loss: 0.0102, MAE: 0.0749
Model: ARIMA Model - MSE: 0.0052, RMSE: 0.0724, MAE: 0.0545

Model with the lowest MAE: Baseline - MAE: 0.014926760457456112
Model with the highest MAE: Multi-step Linear - MAE: 0.21530325710773468

Configuration: SWC_5 - 24 input / 6 output
Model: Baseline - Loss: 0.0397, MAE: 0.0729
Model: Multi-step Linear - Loss: 0.0216, MAE: 0.0434
Model: Multi-step Dense - Loss: 0.0224, MAE: 0.0717

Model: CNN - Loss: 0.0248, MAE: 0.0811
Model: RNN - Loss: 0.0192, MAE: 0.0495
Model: LSTM - Loss: 0.0238, MAE: 0.0749
Model: Autoregressive - Loss: 0.0496, MAE: 0.1197
Model: Bi-LSTM - Loss: 0.0613, MAE: 0.1931
Model: ARIMA Model - MSE: 0.0051, RMSE: 0.0711, MAE: 0.0529

Model with the lowest MAE: Multi-step Linear - MAE: 0.043384749442338943
Model with the highest MAE: Bi-LSTM - MAE: 0.19311939179897308

Configuration: SWC_5 - 48 input / 12 output
Model: Baseline - Loss: 0.0687, MAE: 0.1070
Model: Multi-step Linear - Loss: 0.0379, MAE: 0.0693
Model: Multi-step Dense - Loss: 0.0385, MAE: 0.0927
Model: CNN - Loss: 0.0358, MAE: 0.0831
Model: RNN - Loss: 0.0359, MAE: 0.0888
Model: LSTM - Loss: 0.0423, MAE: 0.1168
Model: Autoregressive - Loss: 0.0691, MAE: 0.1432
Model: Bi-LSTM - Loss: 0.0561, MAE: 0.1443
Model: ARIMA Model - MSE: 0.0050, RMSE: 0.0711, MAE: 0.0528

Model with the lowest MAE: Multi-step Linear - MAE: 0.06925001740455627
Model with the highest MAE: Bi-LSTM - MAE: 0.14426426589488983

Configuration: SWC_5 - 168 input / 24 output
Model: Baseline - Loss: 0.0887, MAE: 0.1009
Model: Multi-step Linear - Loss: 0.0570, MAE: 0.0883
Model: Multi-step Dense - Loss: 0.0572, MAE: 0.1144
Model: CNN - Loss: 0.0562, MAE: 0.1097
Model: RNN - Loss: 0.0569, MAE: 0.1077
Model: LSTM - Loss: 0.0702, MAE: 0.1512
Model: Autoregressive - Loss: 0.2048, MAE: 0.3633
Model: Bi-LSTM - Loss: 0.1434, MAE: 0.2419
Model: ARIMA Model - MSE: 0.0051, RMSE: 0.0716, MAE: 0.0536

Model with the lowest MAE: Multi-step Linear - MAE: 0.08833001554012299
Model with the highest MAE: Autoregressive - MAE: 0.36326563358306885

Configuration: SWC_5 - 168 input / 48 output
Model: Baseline - Loss: 0.1613, MAE: 0.1692

Model: Multi-step Linear - Loss: 0.0878, MAE: 0.1309
Model: Multi-step Dense - Loss: 0.0879, MAE: 0.1541
Model: CNN - Loss: 0.1157, MAE: 0.2234
Model: RNN - Loss: 0.0957, MAE: 0.1677
Model: LSTM - Loss: 0.1393, MAE: 0.2019
Model: Autoregressive - Loss: 0.2286, MAE: 0.3637
Model: Bi-LSTM - Loss: nan, MAE: nan
Model: ARIMA Model - MSE: 0.0048, RMSE: 0.0691, MAE: 0.0505

Model with the lowest MAE: Multi-step Linear - MAE: 0.1309162676334381
Model with the highest MAE: Autoregressive - MAE: 0.36371156573295593

Configuration: SWC_10 - 24 input / 1 output
Model: Baseline - Loss: 0.1040, MAE: 0.2647
Model: Multi-step Linear - Loss: 0.0025, MAE: 0.0158
Model: Multi-step Dense - Loss: 0.0028, MAE: 0.0272
Model: CNN - Loss: 0.0082, MAE: 0.0706
Model: RNN - Loss: 0.0024, MAE: 0.0270
Model: LSTM - Loss: 0.0072, MAE: 0.0613
Model: Autoregressive - Loss: 0.0412, MAE: 0.1407
Model: Bi-LSTM - Loss: 0.0134, MAE: 0.0887
Model: ARIMA Model - MSE: 0.0064, RMSE: 0.0802, MAE: 0.0601

Model with the lowest MAE: Multi-step Linear - MAE: 0.015785953029990196
Model with the highest MAE: Baseline - MAE: 0.26465848088264465

Configuration: SWC_10 - 24 input / 6 output
Model: Baseline - Loss: 0.1236, MAE: 0.2760
Model: Multi-step Linear - Loss: 0.0136, MAE: 0.0425
Model: Multi-step Dense - Loss: 0.0146, MAE: 0.0553
Model: CNN - Loss: 0.0153, MAE: 0.0592
Model: RNN - Loss: 0.0134, MAE: 0.0562
Model: LSTM - Loss: 0.0157, MAE: 0.0608
Model: Autoregressive - Loss: 0.0482, MAE: 0.1538
Model: Bi-LSTM - Loss: 0.0217, MAE: 0.0927
Model: ARIMA Model - MSE: 0.0062, RMSE: 0.0789, MAE: 0.0586

Model with the lowest MAE: Multi-step Linear - MAE: 0.042529407888650894
Model with the highest MAE: Baseline - MAE: 0.2760375738143921

Configuration: SWC_10 - 48 input / 12 output
Model: Baseline - Loss: 0.1427, MAE: 0.2863
Model: Multi-step Linear - Loss: 0.0240, MAE: 0.0511
Model: Multi-step Dense - Loss: 0.0260, MAE: 0.0761
Model: CNN - Loss: 0.0287, MAE: 0.0893
Model: RNN - Loss: 0.0272, MAE: 0.0796
Model: LSTM - Loss: 0.0485, MAE: 0.1489
Model: Autoregressive - Loss: 0.0536, MAE: 0.1430
Model: Bi-LSTM - Loss: 0.0411, MAE: 0.1402
Model: ARIMA Model - MSE: 0.0062, RMSE: 0.0788, MAE: 0.0585

Model with the lowest MAE: Multi-step Linear - MAE: 0.05107926204800606
Model with the highest MAE: Baseline - MAE: 0.2863081395626068

Configuration: SWC_10 - 168 input / 24 output
Model: Baseline - Loss: 0.1483, MAE: 0.2910
Model: Multi-step Linear - Loss: 0.0374, MAE: 0.0848
Model: Multi-step Dense - Loss: 0.0562, MAE: 0.1456
Model: CNN - Loss: 0.0392, MAE: 0.0904
Model: RNN - Loss: 0.0424, MAE: 0.1088
Model: LSTM - Loss: 0.0719, MAE: 0.1918
Model: Autoregressive - Loss: 0.1494, MAE: 0.2999
Model: Bi-LSTM - Loss: 0.0550, MAE: 0.1500
Model: ARIMA Model - MSE: 0.0063, RMSE: 0.0791, MAE: 0.0588

Model with the lowest MAE: Multi-step Linear - MAE: 0.08480609953403473
Model with the highest MAE: Autoregressive - MAE: 0.29994145035743713

Configuration: SWC_10 - 168 input / 48 output
Model: Baseline - Loss: 0.1921, MAE: 0.3152
Model: Multi-step Linear - Loss: 0.0631, MAE: 0.1343
Model: Multi-step Dense - Loss: 0.0782, MAE: 0.1772
Model: CNN - Loss: 0.0810, MAE: 0.1884
Model: RNN - Loss: 0.1044, MAE: 0.2251
Model: LSTM - Loss: 0.1320, MAE: 0.2587
Model: Autoregressive - Loss: 0.2512, MAE: 0.4215
Model: Bi-LSTM - Loss: 0.0790, MAE: 0.1605
Model: ARIMA Model - MSE: 0.0060, RMSE: 0.0773, MAE: 0.0568

Model with the lowest MAE: Multi-step Linear - MAE: 0.13434605300426483

Model with the highest MAE: Autoregressive - MAE: 0.4214521050453186

Configuration: SWC_20 - 24 input / 1 output
Model: Baseline - Loss: 0.2682, MAE: 0.3956
Model: Multi-step Linear - Loss: 0.0010, MAE: 0.0084
Model: Multi-step Dense - Loss: 0.0055, MAE: 0.0260
Model: CNN - Loss: 0.0022, MAE: 0.0307
Model: RNN - Loss: 0.0056, MAE: 0.0534
Model: LSTM - Loss: 0.0098, MAE: 0.0661
Model: Autoregressive - Loss: 0.0158, MAE: 0.0713
Model: Bi-LSTM - Loss: 0.0115, MAE: 0.0839
Model: ARIMA Model - MSE: 0.0096, RMSE: 0.0979, MAE: 0.0693

Model with the lowest MAE: Multi-step Linear - MAE: 0.008398319594562054
Model with the highest MAE: Baseline - MAE: 0.3956223130226135

Configuration: SWC_20 - 24 input / 6 output
Model: Baseline - Loss: 0.2612, MAE: 0.3931
Model: Multi-step Linear - Loss: 0.0067, MAE: 0.0224
Model: Multi-step Dense - Loss: 0.0071, MAE: 0.0359
Model: CNN - Loss: 0.0117, MAE: 0.0688
Model: RNN - Loss: 0.0307, MAE: 0.1416
Model: LSTM - Loss: 0.0169, MAE: 0.0821
Model: Autoregressive - Loss: 0.0173, MAE: 0.0629
Model: Bi-LSTM - Loss: 0.0140, MAE: 0.0761
Model: ARIMA Model - MSE: 0.0095, RMSE: 0.0973, MAE: 0.0685

Model with the lowest MAE: Multi-step Linear - MAE: 0.022449851036071777
Model with the highest MAE: Baseline - MAE: 0.39310890436172485

Configuration: SWC_20 - 48 input / 12 output
Model: Baseline - Loss: 0.2558, MAE: 0.3905
Model: Multi-step Linear - Loss: 0.0123, MAE: 0.0368
Model: Multi-step Dense - Loss: 0.0254, MAE: 0.1146
Model: CNN - Loss: 0.0123, MAE: 0.0457
Model: RNN - Loss: 0.0246, MAE: 0.1017
Model: LSTM - Loss: 0.0241, MAE: 0.0951
Model: Autoregressive - Loss: 0.0375, MAE: 0.1422
Model: Bi-LSTM - Loss: 0.0247, MAE: 0.1043
Model: ARIMA Model - MSE: 0.0095, RMSE: 0.0973, MAE: 0.0685

Model with the lowest MAE: Multi-step Linear - MAE: 0.03683025389909744
Model with the highest MAE: Baseline - MAE: 0.39054426550865173

Configuration: SWC_20 - 168 input / 24 output
Model: Baseline - Loss: 0.2384, MAE: 0.3821
Model: Multi-step Linear - Loss: 0.0200, MAE: 0.0572
Model: Multi-step Dense - Loss: 0.0227, MAE: 0.0832
Model: CNN - Loss: 0.0226, MAE: 0.0819
Model: RNN - Loss: 0.0353, MAE: 0.1144
Model: LSTM - Loss: 0.0442, MAE: 0.1378
Model: Autoregressive - Loss: 0.1201, MAE: 0.2813
Model: Bi-LSTM - Loss: 0.0421, MAE: 0.1509
Model: ARIMA Model - MSE: 0.0099, RMSE: 0.0992, MAE: 0.0709

Model with the lowest MAE: Multi-step Linear - MAE: 0.05719339847564697
Model with the highest MAE: Baseline - MAE: 0.3821398913860321

Configuration: SWC_20 - 168 input / 48 output
Model: Baseline - Loss: 0.2358, MAE: 0.3835
Model: Multi-step Linear - Loss: 0.0346, MAE: 0.0904
Model: Multi-step Dense - Loss: 0.0769, MAE: 0.2180
Model: CNN - Loss: 0.0379, MAE: 0.1276
Model: RNN - Loss: 0.0750, MAE: 0.1862
Model: LSTM - Loss: 0.1249, MAE: 0.2604
Model: Autoregressive - Loss: 0.1368, MAE: 0.2821
Model: Bi-LSTM - Loss: nan, MAE: nan
Model: ARIMA Model - MSE: 0.0095, RMSE: 0.0972, MAE: 0.0685

Model with the lowest MAE: Multi-step Linear - MAE: 0.09037024527788162
Model with the highest MAE: Baseline - MAE: 0.3834894597530365

Feature Importance Using CNN(SWC_5) on 1 week to predict 2 days

Feature: Ppt, MAE: 0.2559828758239746
Feature: SWC_5, MAE: 0.2517089247703552
Feature: SWC_10, MAE: 0.2490413784980774
Feature: SWC_20, MAE: 0.24913401901721954
Feature: SWC_50, MAE: 0.263754278421402
Feature: T_5, MAE: 0.25934356451034546

Feature: T_10, MAE: 0.25772079825401306
Feature: T_20, MAE: 0.25923392176628113
Feature: T_50, MAE: 0.27150267362594604
Feature: Tair, MAE: 0.25002533197402954
Feature: RH, MAE: 0.2244231402873993
Feature: Srad, MAE: 0.2526884078979492
Feature: Wx, MAE: 0.21552443504333496
Feature: Wy, MAE: 0.2303766906261444
Feature: Day sin, MAE: 0.263790100812912
Feature: Day cos, MAE: 0.2635821998119354
Feature: Year sin, MAE: 0.2570425570011139
Feature: Year cos, MAE: 0.26487159729003906

Most important feature: T_50, MAE: 0.27150267362594604
Least important feature: Wx, MAE: 0.21552443504333496